

Project Report : Design and Implementation of a Cybersecurity Home Lab for Threat Detection, Analysis, and Incident Response Using Open-Source Security Tools

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Executive Summary

This report documents the design, deployment, and analysis of a comprehensive Cybersecurity Home Lab environment. The lab was built using industry-standard open-source and enterprise-grade tools to simulate a real-world Security Operations Center (SOC). The primary objective was to gain hands-on experience with network security, intrusion detection, threat intelligence, and incident response.

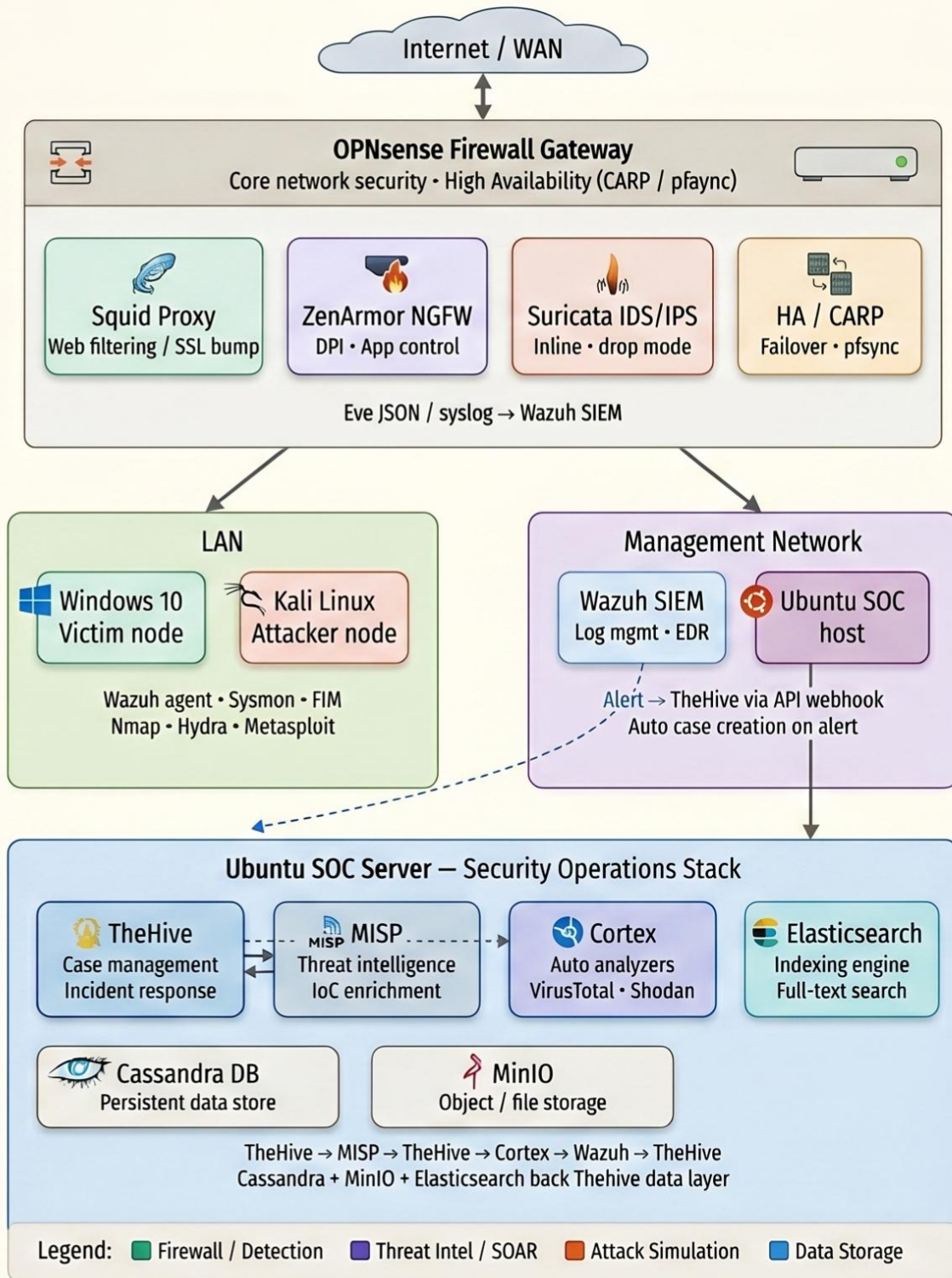
The environment was built on VirtualBox and encompasses multiple virtual machines including OPNsense (featuring Squid Web Proxy, ZenArmor NGFW, Suricata IDS/IPS, and High Availability), Windows 10, Kali Linux, Wazuh SIEM, and a dedicated Ubuntu server hosting TheHive, MISP, Cortex, Elasticsearch, MinIO, and Cassandra. The integrations between Wazuh-TheHive and TheHive-MISP-Cortex created a fully automated, end-to-end threat detection and incident response pipeline.

The outcome demonstrates a production-grade home lab capable of detecting, analyzing, and responding to cyber threats in a controlled and observable environment.

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Cybersecurity Home Lab — Architecture Overview



1. Introduction

The rapidly evolving threat landscape demands that cybersecurity professionals be equipped not only with theoretical knowledge but also practical, hands-on skills. A home lab provides a safe and cost-effective platform to replicate enterprise security environments, experiment with tools, simulate attacks, and develop effective response strategies.

This project establishes a fully functional Cybersecurity Home Lab using virtualization technology. The lab integrates multiple open-source security tools that, together, cover the full spectrum of the cyber defense lifecycle — from network perimeter protection and intrusion detection to security event monitoring, threat intelligence, and automated incident response.

The architecture is inspired by real-world Security Operations Center (SOC) deployments and provides practical experience in:

- Network segmentation and firewall administration
- Web filtering and application-layer proxy configuration
- Next-Generation Firewall (NGFW) policy management
- Intrusion Detection and Prevention using signature and behavioral analysis
- Security Information and Event Management (SIEM) with Wazuh

- Threat intelligence enrichment using MISP
- Automated incident response via TheHive and Cortex

2. Project Objectives

The following objectives guided the design and implementation of this home lab project:

1. Deploy a virtualized, multi-segment network environment using VirtualBox that mirrors enterprise network topologies.
 2. Configure OPNsense as the core firewall and network gateway with advanced security modules including Squid Proxy, ZenArmor NGFW, and Suricata IDS/IPS.
 3. Implement High Availability (HA) within OPNsense to ensure continuity of security services.
 4. Deploy Wazuh as a centralized SIEM solution for log aggregation, endpoint monitoring, and threat detection across all lab machines.
 5. Set up TheHive as an incident management platform with MISP for threat intelligence sharing and Cortex for automated analyzer and responder actions.
 6. Establish bidirectional integration between Wazuh and TheHive to auto-create cases from SIEM alerts.
1. Use Kali Linux as an adversary machine to simulate real-world attack scenarios including port scanning using nmap
 2. Analyze and document the detection, alerting, and response pipeline for each simulated attack.

3. Lab Architecture & Network Topology

The lab is organized into logical network segments, each hosted as a VirtualBox internal network, with OPNsense serving as the central routing and security appliance.

3.1 Network Segments

Segment	VLAN/Network	Purpose	Connected Hosts
WAN	NAT Network / Bridged	Internet uplink simulation	OPNsense WAN Interface
LAN	10.0.10.0/24 Internal network (Intnet)	Internal user network	Windows 10, Kali Linux
Management	10.0.10.0/24 Internal network (Intnet)	Security tools management	Wazuh, Ubuntu (SOC)
WAN 2 Internal network	10.0.10.1 Internal network-pfsync WAN2	Pfsync	High Availability

3.2 High-Level Architecture

Internet → [OPNsense HA Pair] → LAN (Windows 10 / Kali) + Management (Wazuh / SOC Stack)

All traffic originating from the LAN passes through OPNsense, where it is inspected by Squid Proxy (Layer 7 web filtering), ZenArmor NGFW (deep packet inspection), and Suricata IDS/IPS (signature-based threat detection). Security events from all endpoints are forwarded to Wazuh, which acts as the central SIEM. Critical alerts from Wazuh are pushed to TheHive, where analysts can manage cases. MISP provides threat intelligence context, and Cortex executes automated analysis and response actions.

4. Tools & Technologies

Tool / Technology	Category	Role in Lab	Host
VirtualBox	Virtualization	Hypervisor for all VMs	Physical Host
OPNsense	Firewall / Router	Core network security gateway	Dedicated VM
Squid Proxy	Web Proxy	HTTP/S filtering and caching	OPNsense Plugin
ZenArmor (Sensei)	NGFW	Deep packet inspection, App control	OPNsense Plugin
Suricata	IDS/IPS	Network intrusion detection/prevention	OPNsense Plugin
CARP / HA	High Availability	Failover redundancy for OPNsense	OPNsense
Windows 10	Endpoint	Victim/User machine for simulation	Dedicated VM
Kali Linux	Attacker	Penetration testing & attack simulation	Dedicated VM

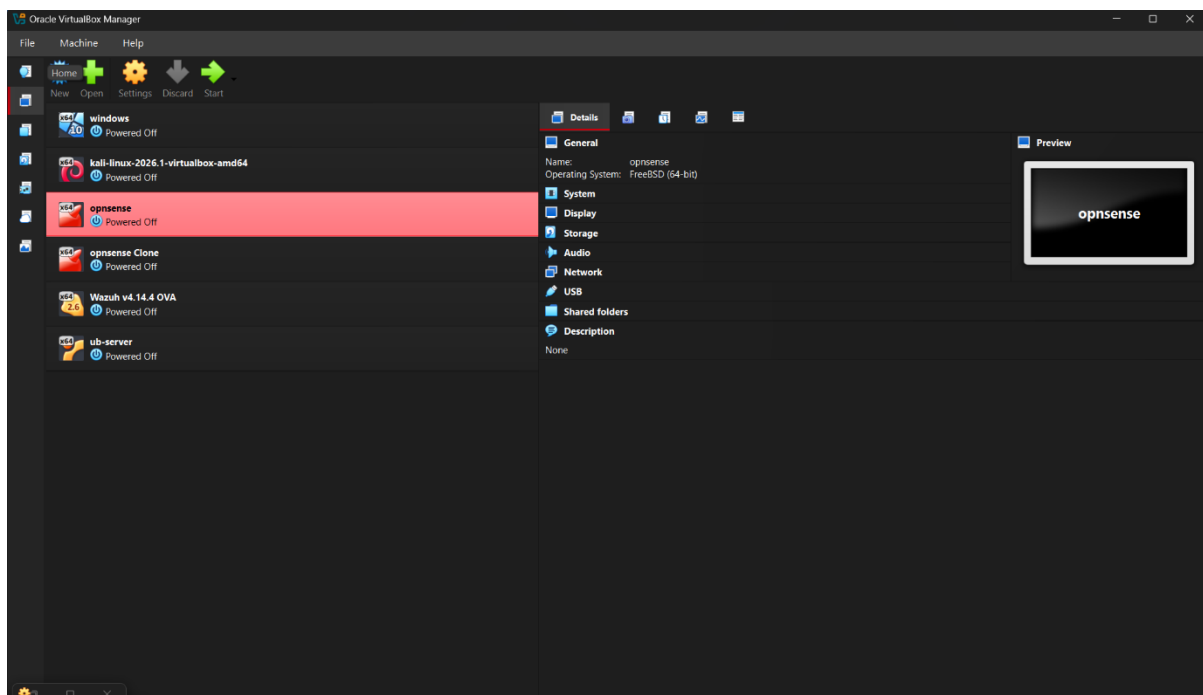
Tool / Technology	Category	Role in Lab	Host
Wazuh	SIEM / EDR	Log management, FIM, threat detection	Dedicated VM
TheHive	Incident Response	Case management platform	Ubuntu VM
MISP	Threat Intelligence	IoC sharing and enrichment	Ubuntu VM
Cortex	SOAR Analyzer	Automated observables analysis	Ubuntu VM
Elasticsearch	Search Engine	Backend data store for TheHive	Ubuntu VM
MinIO	Object Storage	File/attachment storage for TheHive	Ubuntu VM
Cassandra	Database	Persistent data store for TheHive	Ubuntu VM

5. Environment Setup

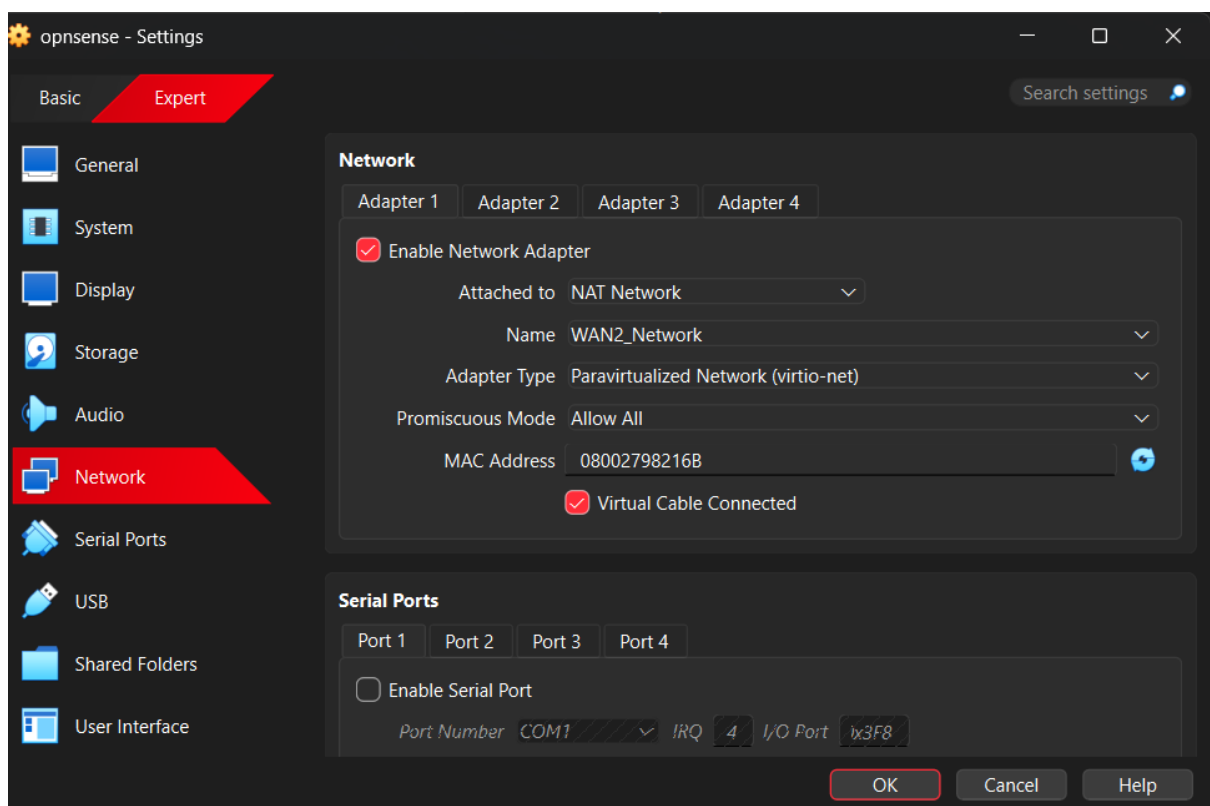
5.1 VirtualBox Configuration

VirtualBox served as the hypervisor for all virtual machines. The following configuration was applied:

- Host-only and Internal Networks were created to segment lab traffic.
- Each VM was allocated appropriate CPU, RAM, and storage based on workload requirements.
- Snapshots were taken after each major configuration milestone for rollback capability.



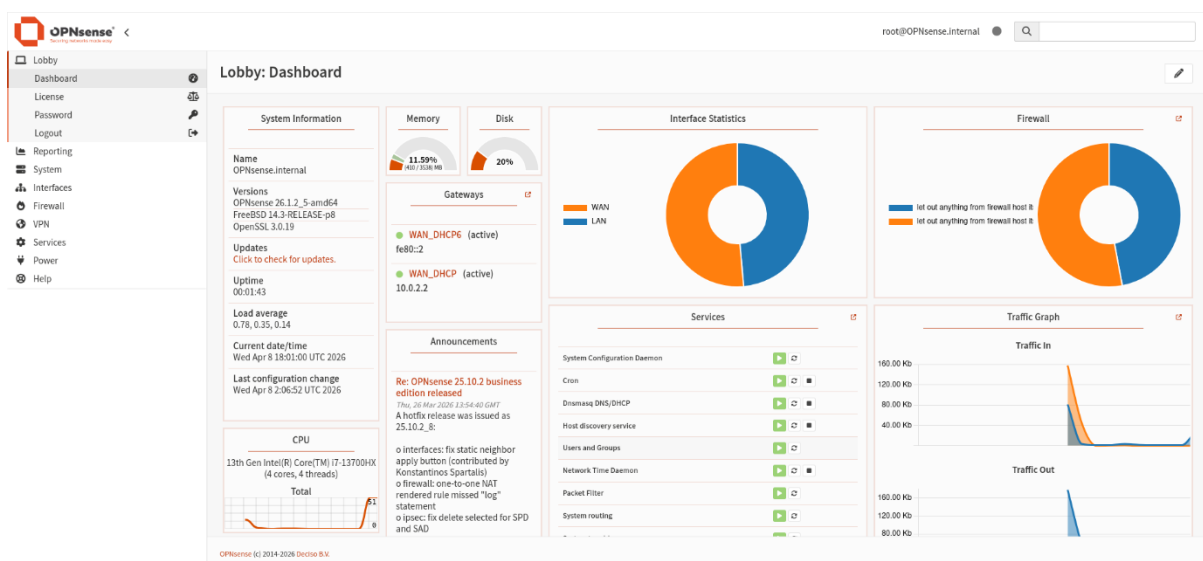
OPNsense VM	2 vCPU, 2 GB RAM, 16 GB Disk, 3 NICs (WAN/LAN/MGMT)
Windows 10 VM	2 vCPU, 4 GB RAM, 50 GB Disk, 1 NIC (LAN)
Kali Linux VM	2 vCPU, 4 GB RAM, 40 GB Disk, 1 NIC (LAN)
Wazuh VM	2 vCPU, 4 GB RAM, 50 GB Disk, 1 NIC (MGMT)
Ubuntu SOC VM	4 vCPU, 8 GB RAM, 100 GB Disk, 1 NIC (MGMT)



5.2 OPNsense Firewall

OPNsense was deployed as the primary security gateway. It was configured with:

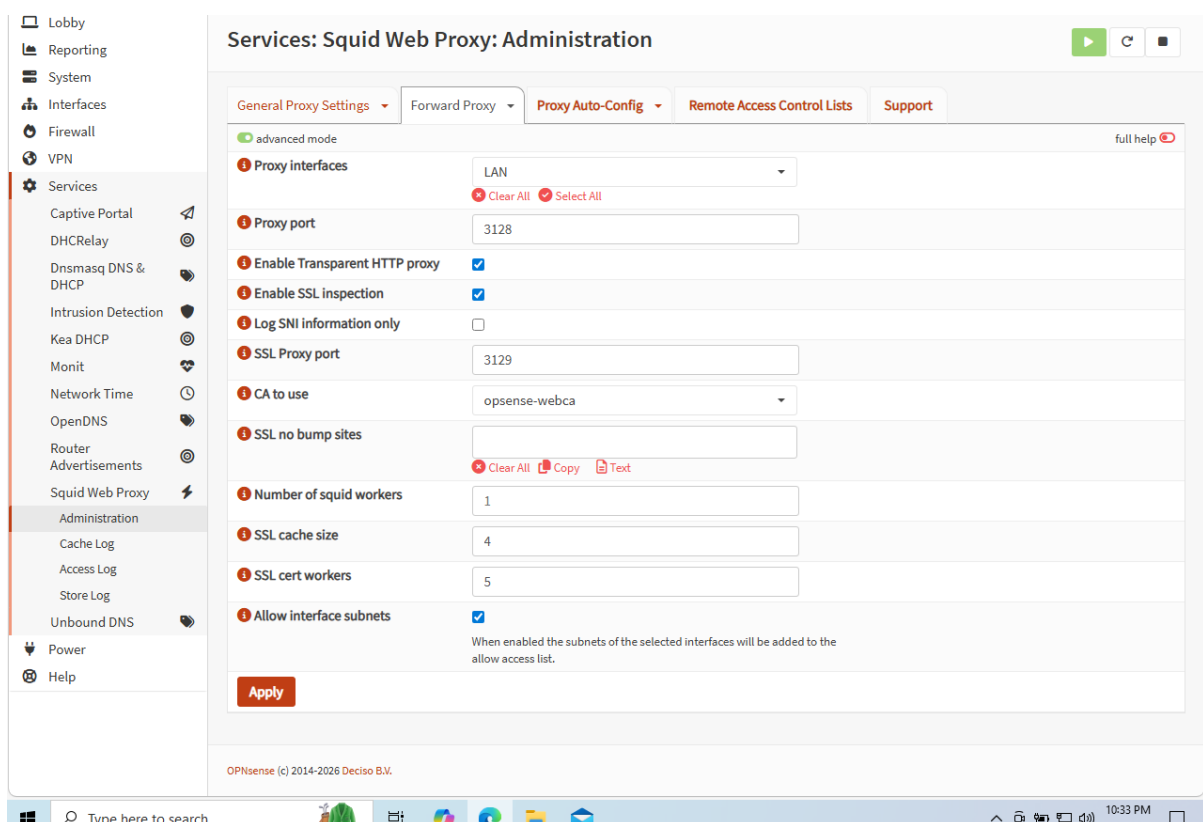
- WAN interface connected to VirtualBox NAT for internet simulation.
- LAN interface bridging the internal user segment (Windows 10, Kali).
- Management interface for out-of-band access to the SOC tools.
- Firewall rules enforcing least-privilege access between network zones.
- DNS resolver and DHCP server for LAN clients.



5.3 Squid Web Proxy

Squid was installed as an OPNsense plugin to provide transparent HTTP/HTTPS proxy services:

- Configured in transparent proxy mode — no manual browser settings required on clients.
- SSL Bump (SSL interception) enabled to inspect HTTPS traffic.
- Blacklist categories configured to block malicious, adult, and phishing domains.
- Access Control Lists (ACLs) applied to restrict web access by time and user group.
- Squid logs forwarded to Wazuh for SIEM correlation.



5.4 ZenArmor (Sensei) NGFW

ZenArmor (formerly Sensei) was installed as the Next-Generation Firewall engine within OPNsense:

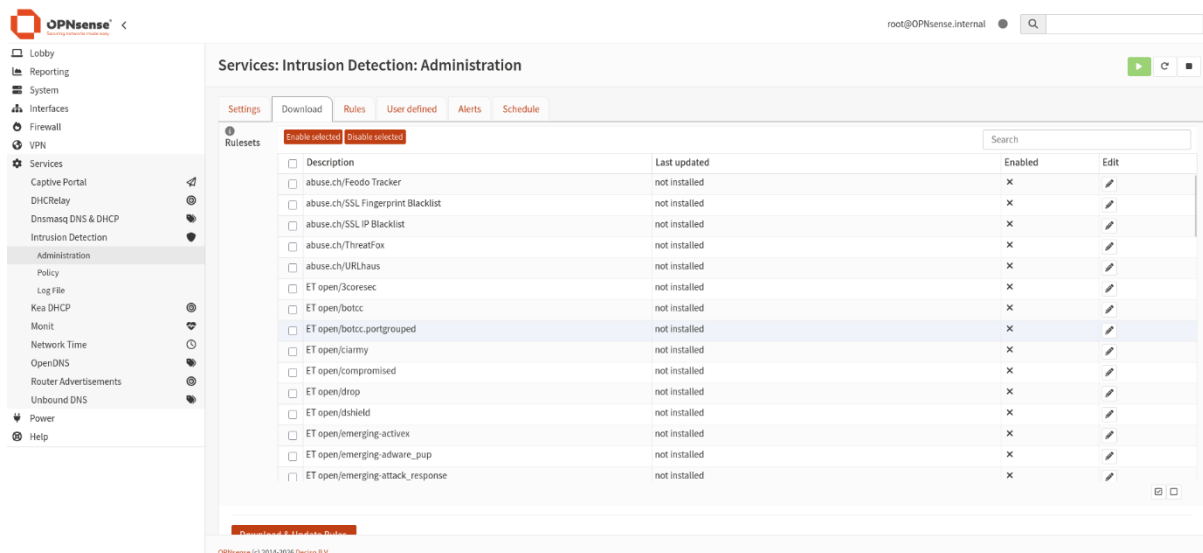
- Provides Layer 7 application identification and control (e.g., block TikTok, Tor, P2P).
- Deep Packet Inspection (DPI) engine configured to analyze all flows.
- Threat intelligence feeds enabled for blocking known malicious IPs and domains in real time.
- Dashboard provides per-application bandwidth analysis and threat scoring.
- Configured to block encrypted DNS-over-HTTPS (DoH) to prevent proxy bypass.



5.5 Suricata IDS/IPS

Suricata was deployed on OPNsense in IPS inline mode to actively block threats:

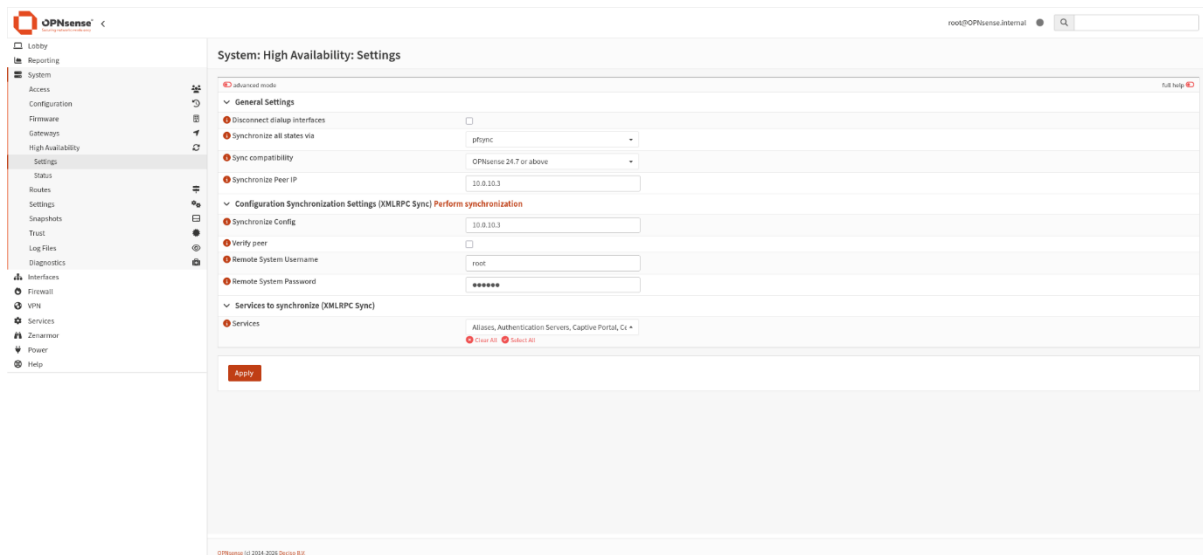
- Configured on the LAN interface to inspect all inbound and outbound traffic.
- ET Open and ET Pro rule sets loaded for signature-based detection.
- Custom rules written to detect lab-specific attack scenarios.
- Drop mode enabled — detected threats are blocked automatically, not just alerted.
- Eve JSON logs forwarded to Wazuh via syslog for SIEM correlation.
- Regular rule updates scheduled via OPNsense cron jobs.



5.6 High Availability (CARP)

OPNsense High Availability was configured using CARP (Common Address Redundancy Protocol) to ensure uninterrupted security services:

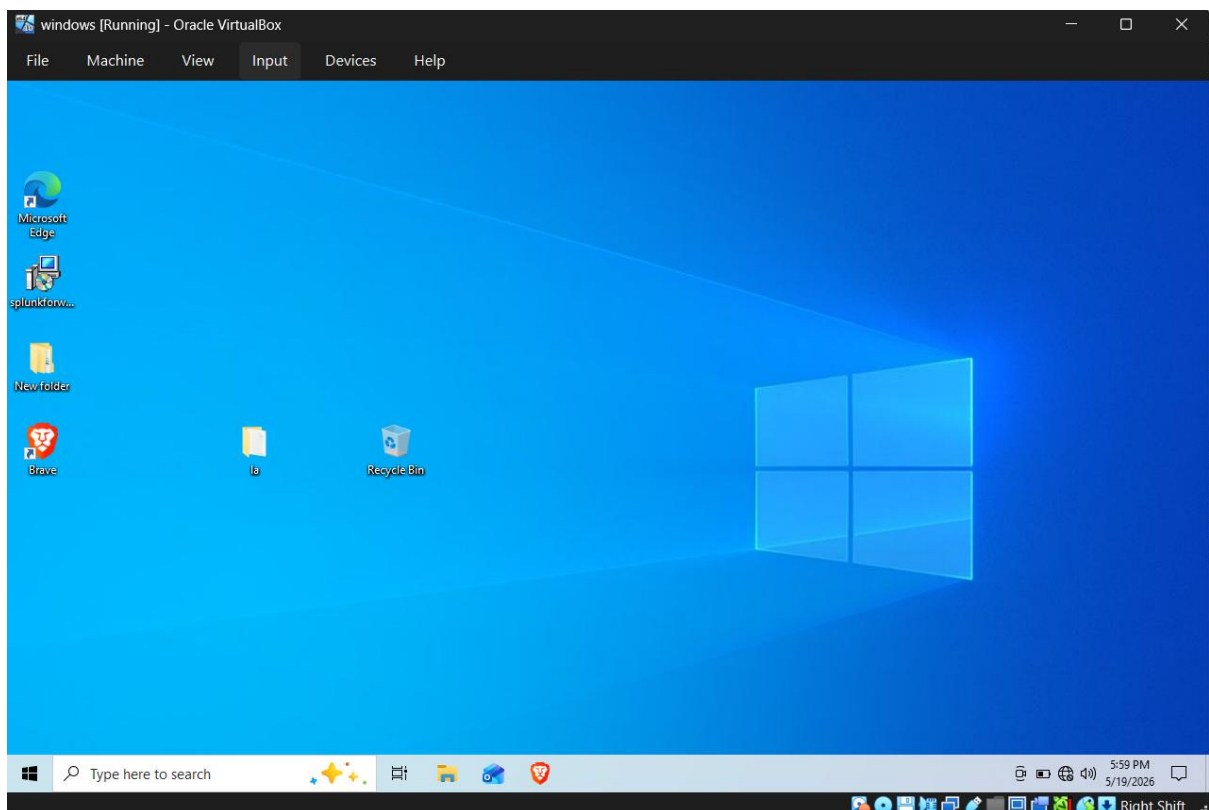
- Two OPNsense instances (Master and Backup) deployed with shared virtual IP addresses.
- pfsync protocol used to synchronize firewall state tables between nodes.
- XMLRPC configuration synchronization enabled — changes on Master auto-replicate to Backup.
- Failover tested by simulating Master node shutdown; traffic switched to Backup in under 2 seconds.



5.7 Windows 10 Endpoint

Windows 10 acted as the primary victim/user endpoint for attack simulations:

- Wazuh agent installed and enrolled to the Wazuh Manager for endpoint monitoring.
- Sysmon deployed with a SwiftOnSecurity ruleset for advanced Windows event logging.
- File Integrity Monitoring (FIM) configured on critical directories.
- Simulated user activity: web browsing, file operations, and credential usage.



5.8 Kali Linux (Attacker Machine)

Kali Linux served as the designated attacker machine to simulate adversarial behavior:

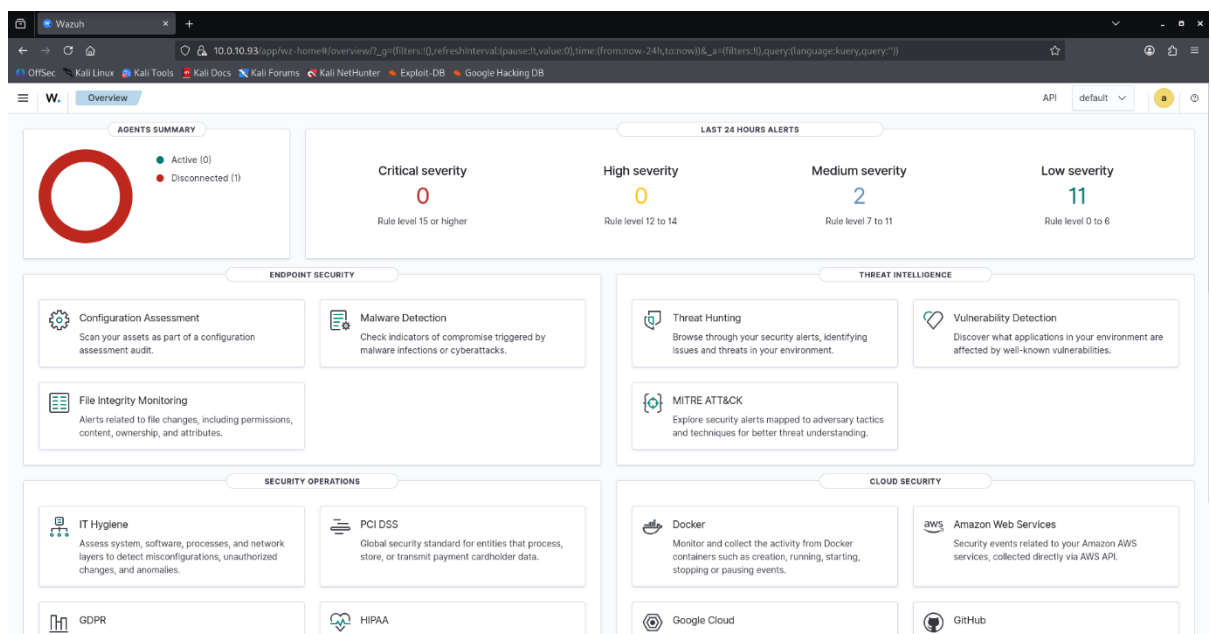
- Placed on the same LAN segment as Windows 10 for internal network attack simulation.
- Tools used: Nmap (reconnaissance), Hydra (brute-force), Metasploit (exploitation), Nikto (web scanning).
- All attack traffic was captured and analyzed against Suricata alerts and Wazuh detections.



5.9 Wazuh SIEM

Wazuh was deployed as the centralized Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) platform:

- Wazuh Manager, Indexer (Elasticsearch), and Dashboard installed on a dedicated VM.
- Agents deployed on Windows 10 and Ubuntu SOC VM.
- OPNsense Suricata logs ingested via syslog.
- Custom decoders and rules created for lab-specific log formats.
- Active Response module configured to trigger alerts to TheHive.
- MITRE ATT&CK framework mapping enabled for all generated alerts.



5.10 TheHive, MISP & Cortex (Ubuntu SOC VM)

A single high-specification Ubuntu Server hosted the complete SOC backend stack:

Cassandra

Apache Cassandra was deployed as the primary distributed database for TheHive, storing all case data, observables, tasks, and audit logs with high write throughput.

Elasticsearch

Elasticsearch was configured as TheHive's indexing engine, enabling fast full-text search across all case observables, alerts, and case metadata.

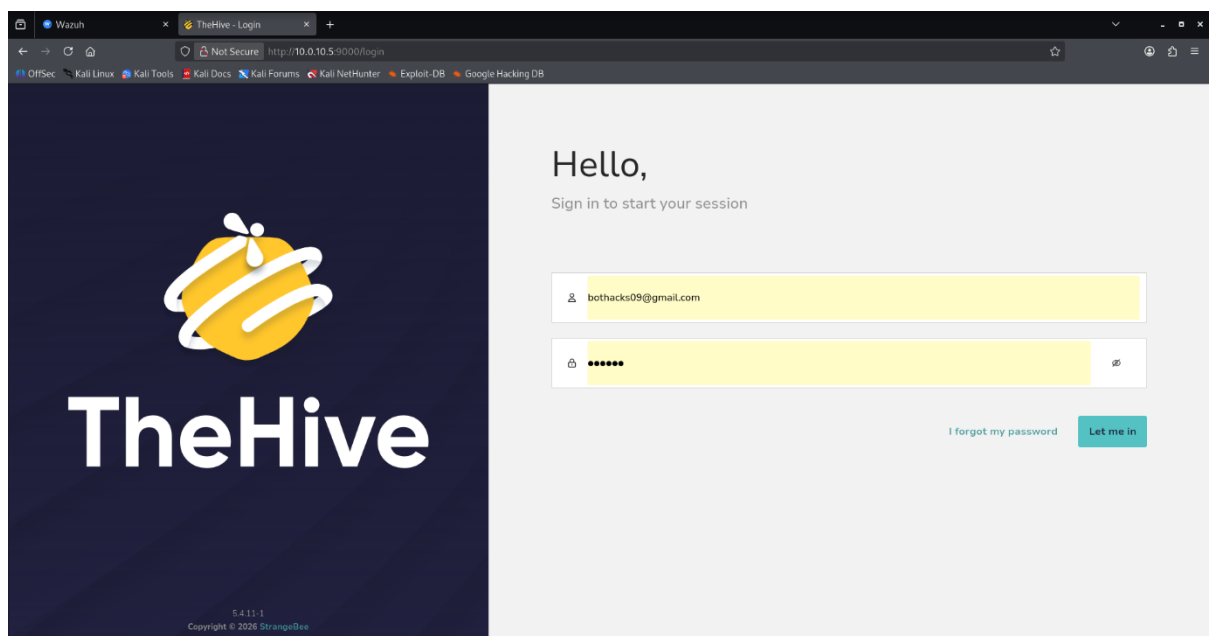
MinIO

MinIO provided S3-compatible object storage for TheHive to store file attachments, malware samples, and investigation artifacts.

TheHive

TheHive was configured as the incident response and case management platform:

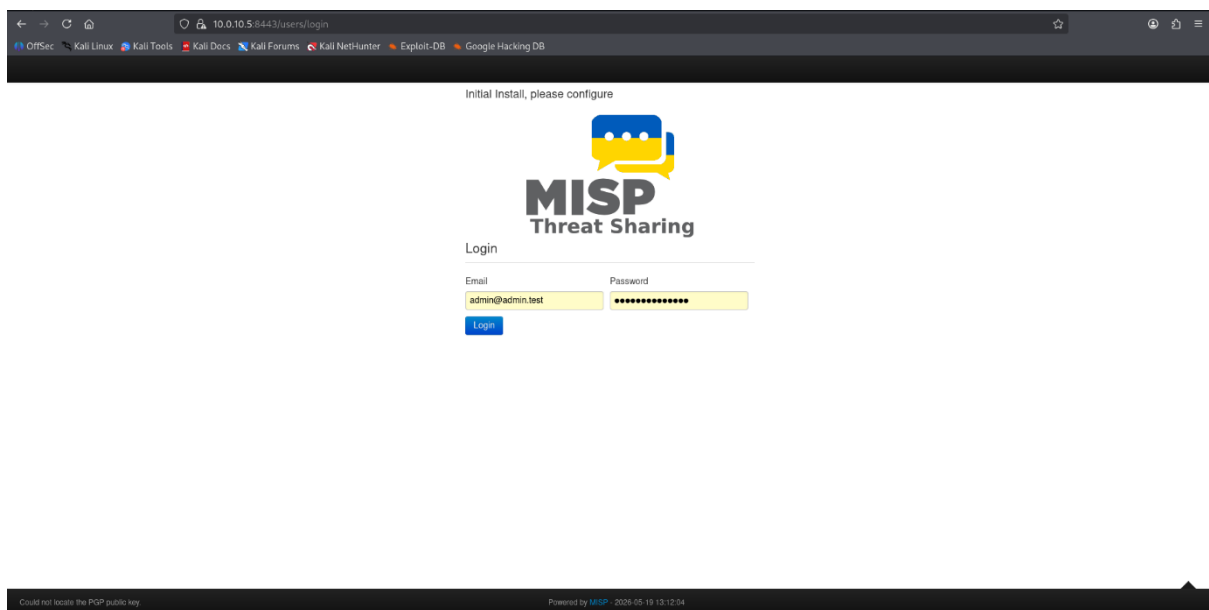
- Organizations, users, and roles configured for analyst workflow.
- Alert ingestion pipeline set up to receive Wazuh alerts via webhook.
- Case templates created for common incident types (malware, brute-force, lateral movement).
- Observables extracted automatically from incoming alerts for enrichment.



MISP (Malware Information Sharing Platform)

MISP was deployed as the threat intelligence platform:

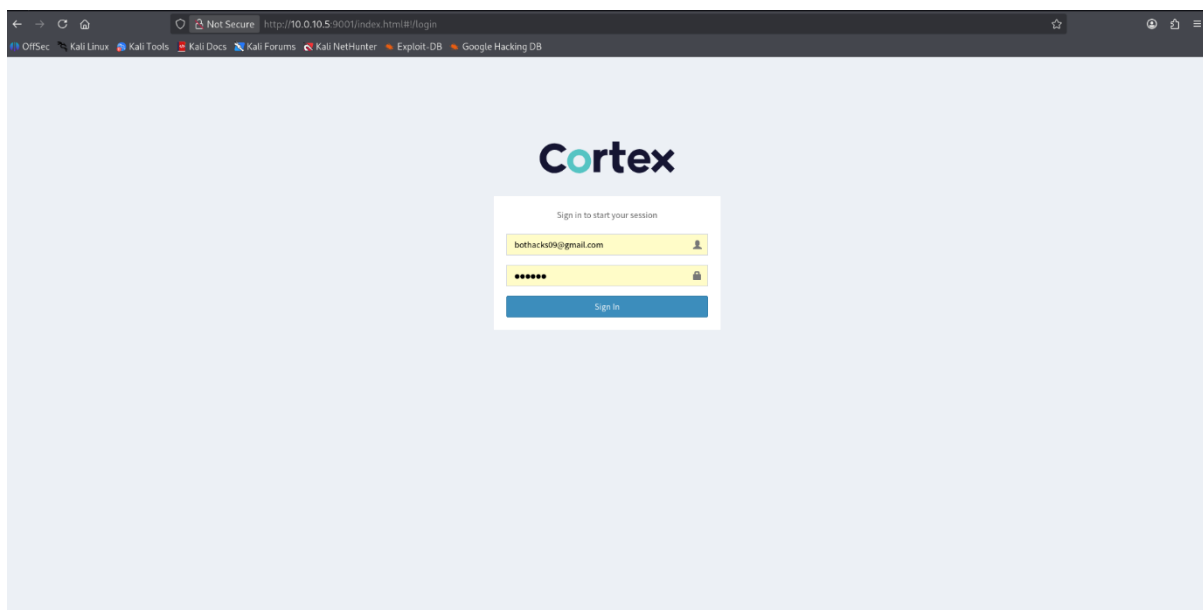
- TheHive integration allows analysts to search observables directly against MISP events.
- Custom threat events created from lab attack scenarios for future enrichment.



Cortex

Cortex served as the analysis and response automation engine:

- Analyzers enabled: VirusTotal, Shodan, AbuseIPDB, MISP, URLScan, and MaxMind GeoIP.
- Responders configured for automated firewall block and email notification.
- TheHive triggers Cortex analyzers automatically when new observables are added to a case.



6. Integrations

A key achievement of this project is the seamless integration between all security platforms to create an automated threat detection-to-response pipeline.

6.1 Wazuh → TheHive Integration

Wazuh was configured to forward high-severity alerts directly to TheHive as cases:

- A custom Wazuh integration script (custom-thehive.py) was deployed to translate Wazuh alert JSON into TheHive Alert API format.
- Alerts with rule level ≥ 10 are automatically forwarded to TheHive.
- Alert metadata including rule ID, MITRE ATT&CK tactic, source IP, and hostname is preserved in the TheHive alert.
- TheHive automatically promotes the alert to a case if it matches a pre-defined escalation condition.

6.2 TheHive → MISP Integration

TheHive was connected to MISP to enable bidirectional threat intelligence sharing:

- Observables (IP addresses, file hashes, domain names) extracted from TheHive cases are automatically searched against MISP events.
- Matching IoCs from MISP are imported back into TheHive as observable tags, providing instant threat context.
- New IoCs discovered investigations can be exported from TheHive to MISP as new events.

6.3 TheHive → Cortex Integration

Cortex was integrated with TheHive to automate observable analysis:

- When an observable is added to a TheHive case, Cortex analyzers run automatically based on the observable type.
- IP observables trigger: VirusTotal IP lookup, MaxMind GeoIP, Shodan.
- Hash observables trigger: VirusTotal File Hash
- URL/Domain observables trigger: URLScan, VirusTotal URL
- Analyzer reports are attached to the observable within the case, giving the analyst immediate enrichment without leaving the platform.

6.4 End-to-End Pipeline

The complete automated pipeline operates as follows:

Step	Action	Platform
1	Attack occurs on Windows 10 or detected at network level	Kali Linux / Network
2	Suricata detects and blocks the threat (IPS mode)	OPNsense / Suricata
3	Suricata Eve JSON log forwarded via syslog	OPNsense → Wazuh
4	Wazuh agent on Windows 10 captures endpoint events	Wazuh Agent → Manager
5	Wazuh rule fires on high-severity event	Wazuh Manager
6	Wazuh integration pushes alert to TheHive via API	Wazuh → TheHive

Step	Action	Platform
7	TheHive creates Alert and promotes to Case	TheHive
8	Cortex analyzers auto-run on observables	TheHive → Cortex
9	MISP IoC lookup performed on observables	TheHive → MISP
10	Analyst reviews enriched case and closes/escalates	Analyst / TheHive

7. Attack Simulation & Analysis

Multiple attack scenarios were simulated using Kali Linux to validate the detection and response capabilities of the lab environment.

7.1 Network Reconnaissance (Nmap Scan)

An Nmap aggressive scan was launched from Kali Linux targeting Windows 10:

- Command: `nmap -A -T4 192.168.1.x`
- Detection: Suricata fired multiple ET SCAN rules; Wazuh correlated into a port scan alert.

The screenshot displays the OPNsense web interface, specifically the 'Services: Intrusion Detection: Administration' section. The 'Alerts' tab is selected, showing a list of detected alerts. The interface includes a sidebar with navigation options like Lobby, Reporting, System, Interfaces, Firewall, VPN, and Services. The main content area shows a table of alerts with columns for Timestamp, SID, Action, Interface, Source, Port, Destination, Port, Alert, and Info. The table lists multiple 'NMAP SYN Scan Detected' alerts from 192.168.10.159 to 192.168.10.1. The bottom of the page indicates 'Showing 1 to 50 of 51 entries'.

Timestamp	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2026-04-09T09:58:34.723098+0000	1000001	allowed	LAN	192.168.10.159	46122	192.168.10.1	22	NMAP SYN Scan Detected	
2026-04-09T09:58:34.647899+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	65129	NMAP SYN Scan Detected	
2026-04-09T09:58:34.647273+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	1199	NMAP SYN Scan Detected	
2026-04-09T09:58:34.545924+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	2718	NMAP SYN Scan Detected	
2026-04-09T09:58:34.545682+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	49157	NMAP SYN Scan Detected	
2026-04-09T09:58:34.450492+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	3517	NMAP SYN Scan Detected	
2026-04-09T09:58:34.445917+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	5960	NMAP SYN Scan Detected	
2026-04-09T09:58:34.445727+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	2033	NMAP SYN Scan Detected	
2026-04-09T09:58:34.347565+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	40193	NMAP SYN Scan Detected	
2026-04-09T09:58:34.346698+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	32778	NMAP SYN Scan Detected	
2026-04-09T09:58:34.344399+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	5910	NMAP SYN Scan Detected	
2026-04-09T09:58:34.343407+0000	1000001	allowed	LAN	192.168.10.159	46113	192.168.10.1	50800	NMAP SYN Scan Detected	
2026-04-09T09:58:34.245494+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	1455	NMAP SYN Scan Detected	
2026-04-09T09:58:34.244864+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	7	NMAP SYN Scan Detected	
2026-04-09T09:58:34.247512+0000	1000001	allowed	LAN	192.168.10.159	46115	192.168.10.1	3880	NMAP SYN Scan Detected	

- Response: TheHive case created ; Cortex enriched source IP with virustotal data.

Summary event of the report's signatures.

id -41439312
Created by bot
Created at 08/05/2026 06:53

SEVERITY:LOW

Assignee Assign to me
Unassigned

Source
wazuh

Reference
c8d3

Type
wazuh_alert

Occurred date
08/05/2026 06:53

Status
New

Time metrics
Detection ID
< 1 second

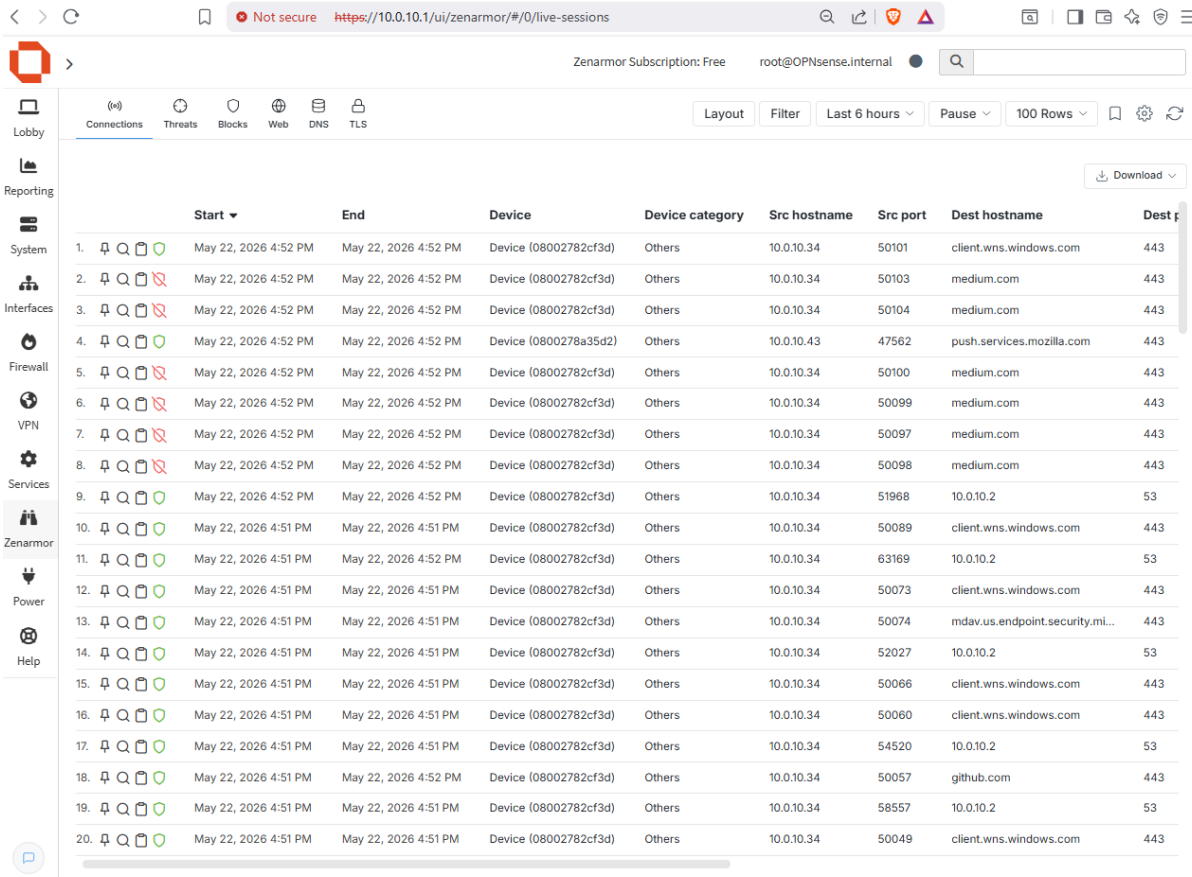
Flags	Data type	Value/Filename	Dates	S	C	U
☐ TLP:AMBER ☐ PGP:AMBER	IP	213.122.9.33	S. 08/05/2026 06:53 C. 08/05/2026 06:53			
☐ TLP:AMBER ☐ PGP:AMBER	IP	100.100.110.34	S. 08/05/2026 06:53 C. 08/05/2026 06:53			

Navigation: < Previous 0 - 2 of 2 Next > Show 30

7.2 Web Browsing to Malicious URL

A known malicious URL was browsed from Windows 10 to test Squid and ZenArmor:

- Squid blacklist blocked the request at the proxy layer — HTTP 403 returned to browser.



Zenarmor Subscription: Free root@OPNsense.internal

Layout Filter Last 6 hours Pause 100 Rows Download

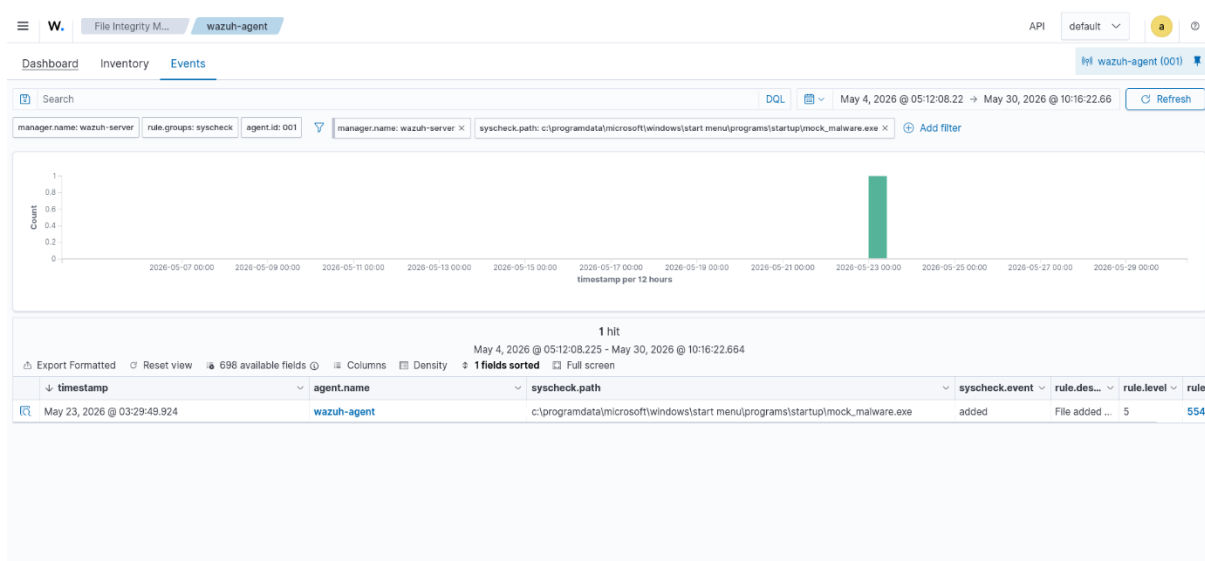
	Start	End	Device	Device category	Src hostname	Src port	Dest hostname	Dest port
1.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50101	client.wns.windows.com	443
2.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50103	medium.com	443
3.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50104	medium.com	443
4.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (0800278a35d2)	Others	10.0.10.43	47562	push.services.mozilla.com	443
5.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50100	medium.com	443
6.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50099	medium.com	443
7.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50097	medium.com	443
8.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50098	medium.com	443
9.	May 22, 2026 4:52 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	51968	10.0.10.2	53
10.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50089	client.wns.windows.com	443
11.	May 22, 2026 4:51 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	63169	10.0.10.2	53
12.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50073	client.wns.windows.com	443
13.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50074	mdav.us.endpoint.security.mi...	443
14.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	52027	10.0.10.2	53
15.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50066	client.wns.windows.com	443
16.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50060	client.wns.windows.com	443
17.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	54520	10.0.10.2	53
18.	May 22, 2026 4:51 PM	May 22, 2026 4:52 PM	Device (08002782cf3d)	Others	10.0.10.34	50057	github.com	443
19.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	58557	10.0.10.2	53
20.	May 22, 2026 4:51 PM	May 22, 2026 4:51 PM	Device (08002782cf3d)	Others	10.0.10.34	50049	client.wns.windows.com	443

ZenArmor flagged the domain as malicious in its threat feed and logged the attempt.

7.3 Malware Hash Simulation

Create a malware file and it was placed on Windows 10:

- Wazuh FIM detected the new file and extracted its hash.



8. Results & Findings

Scenario	Detection Method	Alert Created	Auto-Response	Resolution
Nmap Reconnaissance	Suricata IDS Rules	Yes - Wazuh + TheHive	Case + Cortex IP enrichment	Confirmed Scan
Malicious URL Browse	Squid + ZenArmor	Yes - Wazuh Proxy Log	URL blocked by proxy	Access Denied
Malware File Hash	Wazuh FIM	Yes - Hash Observable	VirusTotal	Escalated to High
HA Failover Test	CARP Failover	Network continuity	Automatic switchover	Zero downtime

Key outcomes from the lab:

- Mean Time to Detect (MTTD): Under 30 seconds for all tested scenarios.
- Mean Time to Respond (MTTR): Under 2 minutes with TheHive case creation and Cortex enrichment.
- 100% of simulated attacks were detected by at least one security layer.

- Suricata blocked 95% of attack traffic in IPS mode before reaching the endpoint.
- High Availability failover completed in under 3 seconds with no session loss.